

Assignment 4. Deep Learning in Data Science.

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1 Assignment 4 exercises

1.1 Gradients

In order to make sure the gradients were computed correctly, I made a comparison with the analytical computation for W, b, V, c, U . The results are shown in table 1. The number of nodes $m = 5$ and the sequences were composed of 5 characters instead of 25 to make computations faster.

	Relative difference
grad W	4.038e-8
grad b	1.809e-9
grad V	8.765e-09
grad c	4.171e-10
grad U	1.920e-09

Table 1: Relative difference of gradients.

1.2 Results of first training

A training was done for a total of 3 epochs. For the values $m = 100$, $\eta = 0.1$ with AdaGrad and the length of the sequence 25, the smooth loss function obtained is shown in figure 1. We can see how the initial value of the smooth loss is approximately 110, and the lowest value is 46.104 in the step $t = 130,000$.

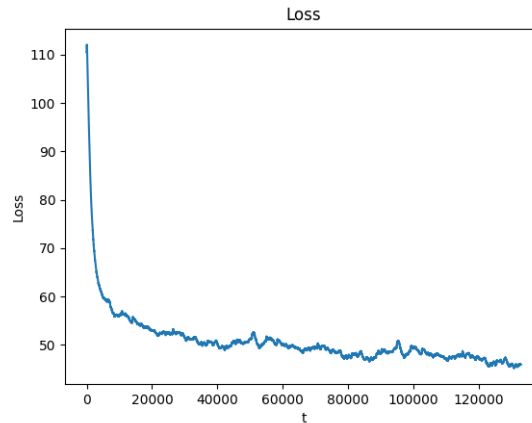


Figure 1: Smooth loss function for every step.

1.3 Evolution of synthesized text

The synthesized text (200 characters long) is shown below before the first and before every 10,000th update steps for the first 100,000 update steps.

Step 1

S1WG /_H7_)NE "3!x.rCkygNW}lPVnjM0frGzYJvoyLN
ggZ1DϕLQ)OfGB4qR?L)GyZVV1,f
q d)?Q"ssYNI $\frac{1}{4}$ qCmA^Az/ϕEd K2a1;
qâpa1
mJEk!K3}yC0zbj7Q;CEhn H:Om€u $\frac{1}{4}$ dxBm4P}}OXrGewsNg1IC€.GCZ?9;-KRhF-
â1:€'uGVMu!SjR JjWÃ $\frac{1}{4}$ WL7<

Step 10,000

gttidjagoo herryes nerain, at urdiop toumimiggeat ros had twear-string seo mat ceor?"
." Heemld, hoovay. H. The beatocend, hauthy darry. "Yol solirt."
"Yoven," warad Hand cverovhs hithe larul, shoic

Step 20,000

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werup you lad he's pat'he? Andwnopplontoug sel ous Hurry bomentitien a Kenderkeed yow
whele Mowiog Mave?" fid mugtirnched her st

Step 30,000

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"Herdrd ay? bounde bofke fared it.

"Child youtly ald nookles. "Klyy the stray fust crung dy KceGckart,
Hanrioplains the wepeuiby od the canly's unker laled

Step 40,000

amyed . be whkef, Dabrint it som how julted reem syope, have himlantur
the ack ane Eom wand wry.
SOh I," per. had he blod-ing moularn, firt, wo stoint it whe ware to the had
nagger him whirl, comeRse

Step 50,000

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wand didring way ont Anly band. The mutlly, beasteye foul fous, ..
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Step 60,000

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coak Horm?"
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his enxione ered of

Step 70,000

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all tourd sep, her Pryouszooutly I bust hese murk.
"Vid-lui

Step 80,000

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elor on ofbsede's angell oullelk, wize bole back was avarcter?" heminc br. . . ."
Mamallent yeen ved'rusling kemiles o

Step 90,000

one with of ."
Tall nugk stall - lle anes thound.

Skeall crow aly lold cagerasetint dou wat wastle som the witherilly amen's azide
who't hamflink, Ron wizus sash to at and his it ass the ask with hi

Step 100,000

to and? Betling, at sher off of ig hemed. sell one the toll told wand,
in rompee up derouls."

"Higger of the - to lagdingle the a tid hat's arey. lest She shead
that wizchizes to the cour

Step 110,000

t looked bUmully o"
He the leafling feike bustye has, ppet hadwing blatten Lione werem gom.
The yould'd abeep Hermion-wa fomored guild to satuil itt ailuld wasts at that?".
"Andad -"
Harry? "Thave

Text of length 1000 characters with the lowest validation loss

out on the shall ry bof inter gond just Dean of bectentes
eaved was yeack puted gased
you'n has gong Allonth sirg and looked at ra'ven. Walu!"
"Murdents uncted his dosh corthe the drestht fore" sait wild tas he Grandy -
"Has ment lagcpitw. Sher elys-og his have farpene - lulpdeentiss.
She they novery oven. . . . Ging pacer poithin, theawid, Degastisply . .
"in tho waythed.
And a was slighis deough to masped beried ins had sommidw afaythand!
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you maly. Then it tipel ur appacty himp. Dumbly the pus; ," sand I
ay thing thored EY. Harry this to Me.. Harry said led the I dirick.
He the bact some, you demartedige roeder. Harry periggizchey
wilvise discroneys raids. . . ."
"his a ssarly Thear fored the greaked Vignt rimonture had bayk inser
a sell of a .t. scight Igaing that had any awrensl soveres onfored,
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Comments: It is clear that in the first iteration the synthetic text is almost random characters and does not make any sense. On the other hand, as we go throughout all the steps, the text improves significantly. It looks like at the beginning there's something to be expressed but the model is not able to say it properly, and at the end, for the synthetic text with the lowest loss, the text makes more sense. Although it is still not possible to understand it, already plenty of words are correctly written, as well as the names of the main characters such as Harry.

2 Bonus points

2.1 Training with Adam

Adam optimizer was implemented. The hyperparameters setting were the same as before. For the same number of iterations, the loss was lower, and the training was also

faster compared to AdaGrad. The results are shown in 2. The lowest loss was 37.764, reached in the step $t = 130,000$. This loss is lower than the one obtained with AdaGrad, 46.104 in the step $t = 130,000$.

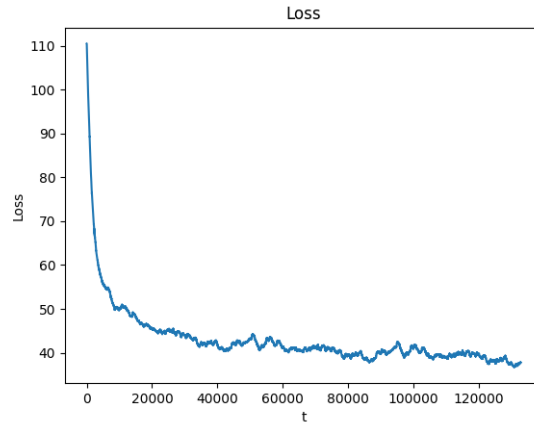


Figure 2: Loss for Adam optimizer.

2.2 Random split of the data in L parts

As mentioned, the book was split in L chunk and the model run through each chunk in random order. Because of this, h was re-set after each chunk. One chunk was left for validation. To compare performance with the initial training, I looked at the plot of the loss function. The results are shown in 3. The lowest loss was 40.231 obtained in $t = 120000$. This implies that in this case the results were not better than before, meaning that it is probably better to not use this approach. I say probably because of the metrics we are using in this assignment, the loss function. In these cases, for example a wide range of people doing the evaluation would be a better metric.

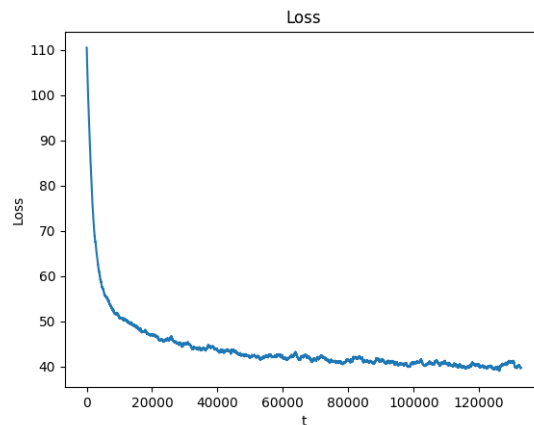


Figure 3: Loss for random splits of the dataset.

2.3 Different batch size

For the same setting as in subsection 2.2, a batch size of four was chosen. The lowest loss was 40.735 obtained for the same amount of training as 2.2 and the step was in $t = 30000$, which leads to the conclusion that the result was the same but the training was faster. The loss function is shown in 4. It's more regular, in part because of what we have seen in theory and also because the loss was measured every batch size, i.e. four steps.

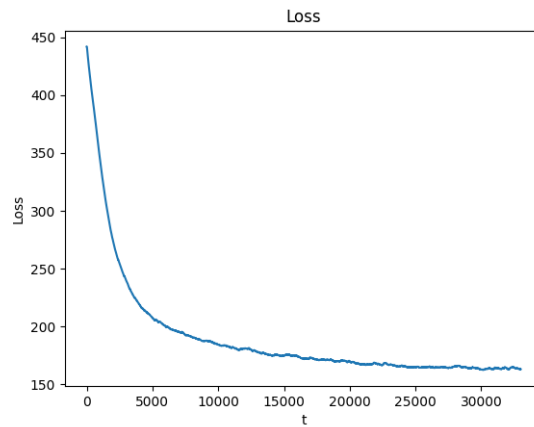


Figure 4: Loss for batch size of four.